

Traffic Flow Prediction Model Using Google Map and LSTM Deep Learning

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Abstract— Traffic jam is the most important factor of the urban road networks for making advanced travel plans, estimating traffic density and proactively managing the traffic flow. It causes adversely affect the social life, country economy, human health and is sometimes unable to manage traffic flow and signal. We explore the stacked long short-term memory (LSTM) network model to perform the multi-step ahead traffic speed prediction by employing Google Maps real-time and historical traffic data of three different types of urban road sections. After that, a Time-dependent correlation algorithm is used to map the predicted speed into the predicted traffic flow. The experimental results explored that, propose stacked LSTM model's multi-step advanced predicted traffic flow mean relative error is varying between 8.25% ~14.09%. Also, results showed that the prediction accuracy improves and is stable with the freeway and identical traffic flow.

Keywords—RNN, LSTM, Google Maps, Intelligent Transportation System, Application Programming Interface, Time-Dependent Correlation, Traffic speed and Traffic Flow.

I. INTRODUCTION

Traffic demand increases in urban road networks over time in Bangladesh; especially in Dhaka and Chittagong. The number of the population increasing day by day and badly affected by massive traffic jams in our daily life. Traffic jam is the main problem of Dhaka city due to the number of traffic, population growth, existing traffic policy, ongoing road construction works, lane violation, unnecessary parking, narrow road capacity, overtaking tendency and lack of intelligent transportation systems (ITS), etc. The traffic jam causes an important portion of working hours left from the travellers, riders, logistics services and drivers which are indirectly put into an adverse impact on the country economic system and social life. Moreover, it's directly impacted the urban roads traffic signal control and managing the traffic flow on daily basis. The transportation engineers (TE) are facing challenges corresponding to traffic congestion and safety issues for increasing traffic demand day by day [1-2] on the urban road networks. This can affect the overall traffic planning and design of the transportation systems, traffic routes, policy-making and developing alternate solutions strategies, etc. To avoid traffic jams and assist transportation management (TM), a traffic flow prediction model could be a better solution for urban road networks. Proactively traffic control systems are of great importance and efficient for transportation management systems [3-4]. Specifically, one-

step or multi-step advanced traffic flow prediction is an important factor for proactively traffic control, travel planning, transportation network planning, reducing fuel cost, time-saving, intelligently route guidance, location-based services and avoiding traffic jams [5-12]. Moreover, the prediction may differ from the actual, due to a range of factors suppose; properties of road networks, traffic mode, traffic policy, vehicle types, driving policy, natural disaster, socio-economic condition, weather condition, the political situation and instantaneous road traffic conditions such as ongoing construction or roadblock, accident, etc [6,13,14]. In the meantime, several researchers have been proposed traffic flow prediction models from various sources of data suppose; roadside cameras, GPS networks, traffic engineering databases, traffic microwave detectors, number plate detection, induction loop detection, sensor networks etc. Moreover, the traffic-related data has been collected from freeways or highway and their everyday traffic data patterns are almost similar. However, these proposed models are difficult to obtain in Bangladesh due to mixed types of vehicles, costly, insufficient traffic database and traffic management networks.

The study is concerned with effectively developing a stacked two-layers LSTM based deep network for traffic flow prediction model using Google Map real-time traffic information and other sources of data. In general, traffic data suppose; traffic speed or traffic flow represent sequential characteristic where future step depends on previous steps. In this context, LSTM and recurrent neural networks (RNN) are more well-known for predicting such sequences of time series data. However, RNN is more suitable for processing short-term sequences due to its inability to maintain cell memory without further transformation. It suffers from a gradient decent problem also the same as a conventional neural network. The LSTM network removes such vanishing gradient problems by employing a memory cell with three gates; input, forget and output. Also, it can learn long-distance information through the sigmoid functions and cell state information stored in the memory for future reproducing. In this study, we propose a two-layer stacked LSTM deep learning model for one-step and multi-step ahead traffic flow prediction using Google Maps traffic information. Google Maps provided data are reliable, hassle-free, easily accessible, low cost and easily extractable 24/7. In addition, Google Map is one of the reliable traffic information in a developing country like Bangladesh. The rest of the paper is organized as follows; II. Methodology, III. Data collection and pre-processing, IV. Model simulation, V. Results and discussions, VI. Conclusion and future work.

II. METHODOLOGY

Traffic speed is a vehicle's rate of motion from one point to another point with a specific direction. Space mean speed is the average speed of all vehicles occupying a given section of a road over a specified period. Simply, we define the space mean speed $\bar{v}_s = \frac{d \text{ (Mile)}}{t \text{ (Hr)}}$ where d = distance travelled or length of roadway segments (km) t = average travel time (hr) of that segments. We used space mean speed as an input for a model. The traffic flow is measured by how many vehicles pass per hour per lane in an intersection. In this study, we developed two-layer stacked LSTM deep learning models for traffic speed prediction using real-time and historical data. The stacked LSTM networks make the model deeper by adding more layers and employing first layer output placed into the next layer LSTM as input. The two-layer stacked LSTM model is defined through hidden cells or neurons as per optimum hidden cell rule, epoch, and batch size defined according to the keras recommendation and standard activation function (relu) for the highest level of prediction accuracy. The model predicted speed is mapped with the developed time-dependent correlation equations. Time-dependent correlation equation developed by single day collected field level traffic flow, traffic speed and timestamp.

A. LSTM Basic Cell Structure and Prediction Procedure

The basic LSTM cell or memory structure is shown in Fig-2.1. The information adds or removes to or from the cell memory through a control unit called a "gate". The LSTM cell consists of three gates – input, forget and output with a sigmoid and relu activation function which is multiplied by the output of other neurons. Using the forget gate, LSTM can selectively remember or forget previous cell information to the current cell memory that passed through an activation function [7]. Similarly, input and output gates control the input and prediction through the rectified linear unit (relu) activation function. Suppose, LSTM receives input x_t at time t when the previous cell output is h_{t-1} and cell state is C_{t-1} .

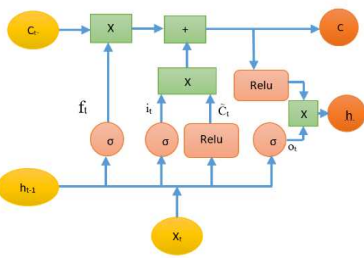


Fig-2.1, LSTM cell structure

$$\begin{aligned} \text{Forget gate,} \quad f_t &= \sigma(U_f \cdot x_t + W_f \cdot h_{t-1} + b_f) & (1) \\ \text{Input gate,} \quad i_t &= \sigma(U_i \cdot x_t + W_i \cdot h_{t-1} + b_i) & (2) \\ \text{Output gate,} \quad o_t &= \sigma(U_o \cdot x_t + W_o \cdot h_{t-1} + b_o) & (3) \\ \text{Cell input state,} \quad \tilde{C}_t &= \text{relu}(U_c \cdot x_t + W_c \cdot h_{t-1} + b_c) & (4) \\ \text{Cell output state} \quad C_t &= f_t * C_{t-1} + i_t * \tilde{C}_t & (5) \end{aligned}$$

Here, $f_t, i_t, o_t, \tilde{C}_t, C_{t-1}$ and C_t are carrying the same dimension. The U and W are weight vectors of forget, input, output gates and cell state, b_f, b_i, b_o, b_c are the input bias of these three gates and the cell input. The symbol σ represents the sigmoid function, $f(x) = 1/(1+e^{-x})$ and relu represents the rectified linear

function $f(x) = \max(0, x)$. The hidden cell output, h_t can be calculated as per Eq. 6.

$$h_t = \text{relu}(C_t) * o_t \quad (6)$$

B. Sequential Data Prediction Using LSTM Networks

The LSTM network is suitable for analyzing sequential or time-series data in various aspects like travel time, traffic speed and traffic flow prediction through different types of models [7,15,16]. The multi-layer LSTM model prediction accuracy higher compare to the single-layer. Two layers stacked LSTM sequential model is shown in Fig-2.2. For

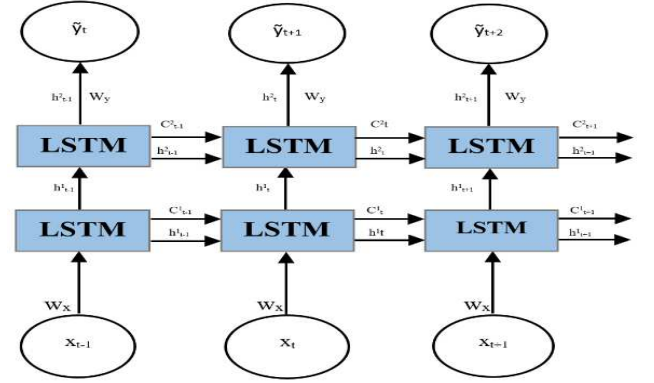


Fig-2.2, Stacked LSTM Sequential Structure

For example, considering at t time, the input of LSTM network is the historical data of traffic speed like x_{t-1}, x_t, x_{t+1} and the predicted output at one-step ahead are $\tilde{y}_t, \tilde{y}_{t+1}$ and $\tilde{y}_{t+2}$ respectively. Two step-ahead prediction input sequences like, $\{x_{t-1}, x_t\}, \{x_t, x_{t+1}\}, \{x_{t+1}, x_{t+2}\}$ and predicted outcomes are $(\tilde{y}_{t+1}), (\tilde{y}_{t+2})$ and $(\tilde{y}_{t+3})$ respectively. Similarly, three-step ahead prediction $\tilde{y}_{t+3}$ can be done by previous three steps $\{x_t, x_{t+1}, x_{t+2}\}$ sequences of observations. The actual output of the stacked LSTM cell will be as per the below equations.

$$\tilde{y}_t = w_y \cdot h_{t-1} + b \quad (7)$$

$$\tilde{y}_{t+1} = w_y \cdot h_t + b \quad (8)$$

$$\tilde{y}_{t+2} = w_y \cdot h_{t+1} + b \quad (9)$$

Where W_y is the weight matrix between output and hidden layer and b is the bias of the output layer.

C. Time-Dependent Correlation Algorithm

Stacked LSTM model predicted traffic speed employed to time-dependent correlation for forecasting the traffic flow for a road section. In this context, traffic speed and traffic flow data are collected for several road sections from Google Maps and actual or field level at the five-minute interval for a day from 8:00 AM to 12 PM and 3:00 PM to 7:00 PM. A time-dependent correlation is developed using traffic speed collected from Google Maps and field level real-time traffic flow according to their timestamp for each road section. Time-dependent correlations between traffic speed and traffic flow have some positive and negative correlations [17]. Every day, traffic speed ranges are limited and the same speed exists multiple times at morning and evening hours whereas traffic flow is different for the same traffic speed. In this situation, six consecutive traffic speed, and field-level traffic flow data are employed for getting strong correlation

equations where co-efficient R square is a positive or strong correlation. The traffic flow calculation is done by using an algorithm developed by time-dependent correlation equations (traffic flow $f=a+bs+cm+dh$) where b, c, d are coefficients of predicted speed, minute, hour and a is the intercept for a road section. The algorithm is mapped traffic flow prediction based on time and predicted traffic speed.

III. DATA COLLECTION AND PRE-PROCESSING

The traffic speed data were collected from three different signalized, non-signalized, and freeway road segments at five minutes intervals from 8:00 AM to 12:00 PM considered morning hour and 3:00 PM to 7:00 PM considered as evening hour whereas each segment length range 140m to 300m. At the same time, weather information and adjacent road speed as well as VIP movement road speed data were collected for getting the highest level of prediction accuracy. These three road segments are Sainik club intersection; CTW (Centre to West) is considered little bit demand and CTN (Centre to North) is high demand traffic flow which is shown in Fig-3.1 and Army stadium road section STN (South to North) is considered as free flow or freeway traffic analysis which is shown in Fig-3.2 an urban road network. Every day, we have collected traffic speed data around 48 instances for morning and 48 instances for evening hour, and the total collected data per day is 96*3 instances for these three intersections. In such a way, the data collection duration was continuous one month. The travel speed, segment, lanes, adjacent road's speed, VIP road speed, and weather information were extracted from Google Maps and respected website by using distance matrix API with necessary python libraries.

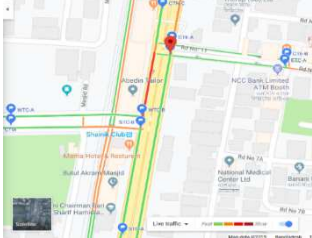


Fig-3.1, Sainik club intersection



Fig-3.2, Army stadium road section

IV. MODEL SIMULATION

The developed model's simulation process is performed by hidden neurons n_h , max epochs 500, the batch size 128 including early-stopping with patience and model checkpoint etc. for higher accuracy. The model's number of hidden neurons are assigned as per below Eq. 10 for optimum accuracy and performance.

$$n_h = N_s / (\alpha * (N_i + N_o)) \quad (10)$$

Where N_s is the number of observed samples, N_i Input features, N_o is the number of output neurons and α is an arbitrary scaling factor usually range 2-10. In this model, we employed an arbitrary scaling factor $\alpha=2$ and got optimum prediction accuracy and minimum error in each road segment. Proper training is the prerequisite for getting better prediction accuracy through the developed LSTM deep network model. Moreover, the optimized parameter setting is an important part of the model train and validation phases. During the simulations period, the model trained and validated the model through Eq. 1-7 for predicting traffic speed. We split train

input samples into 67% and 33% for training and validation. During simulation 2688 and 1344 samples are employed for training and 96 and 48 samples are considered for traffic flow prediction or test. The traffic flow prediction model trained flowchart is shown in Fig-4.1. The proposed model simulated for maximum three-step ahead prediction. The model predicted traffic speed mapped into traffic flow using a time-dependent correlation algorithm. After training the model, three criteria's are used to measure the model prediction error and accuracy as presented in Eq. 11-13. The prediction accuracy measures the mean relative error of the model. A small relative error indicates a high predictive performance of the model.

$$\text{Mean Absolute Error, } \sum^N (|x_{t+1} - \hat{y}_{t+1}|) / N \quad (11)$$

$$\text{Root mean square error, } \{\sum^N (x_{t+1} - \hat{y}_{t+1})^2 / N\}^{1/2} \quad (12)$$

$$\text{Mean Relative Error, } \{\sum^N (|x_{t+1} - \hat{y}_{t+1}| / x_{t+1}) / N\} \quad (13)$$

Where N is the total number of test or predicted samples.

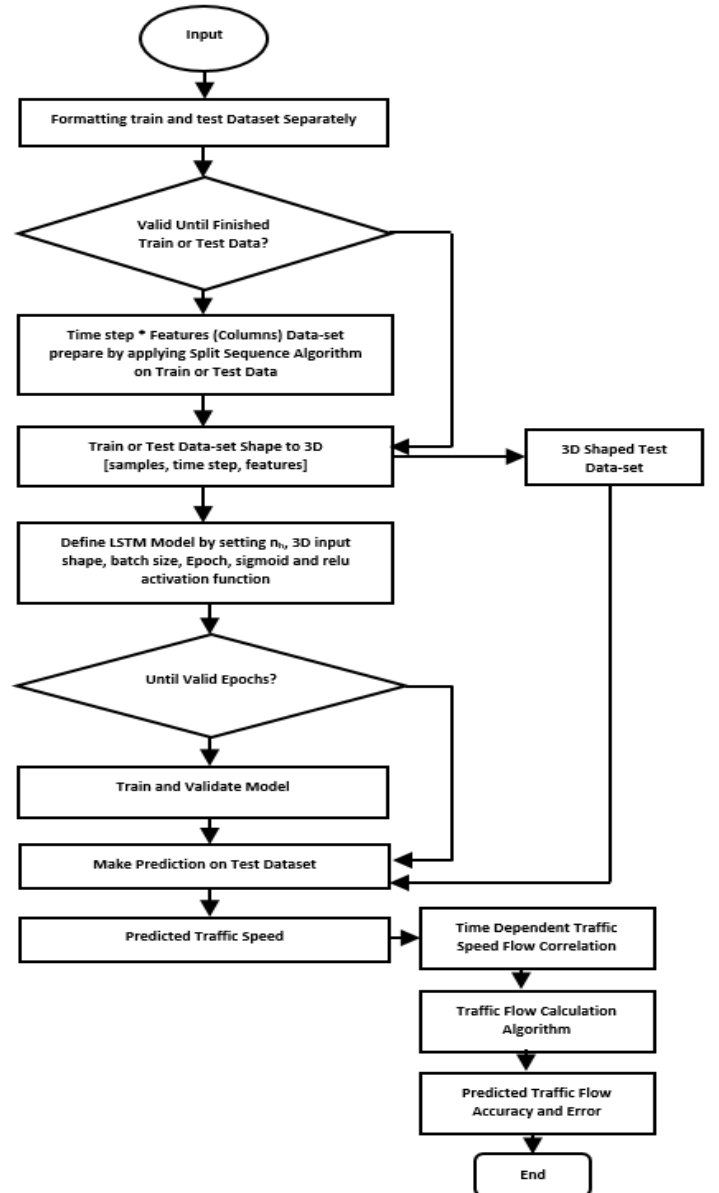


Fig-4.1 Model Training Flow Chart

V. RESULTS AND DISCUSSIONS

To investigate the proposed model traffic flow prediction performance and accuracy, three different types of experiments are performed separately at five minutes interval's dataset. The experiment is performed on three road sections for predicting the multi-step traffic flow through model-1 (Sainik club, Centre to West, CTW), model-2 (Sainik club, Centre to North, CTN) and model-3 (Army stadium, South to North, STN) road section. These models traffic flow prediction performance is analyzed by using one-month historical traffic data. The model-1, CTW road section's non-motorized vehicles (NMV), bus and CNG flows stable at all times whereas motorcycle (MC) and car flow uncertainly change each time. After simulating developed model-1, the predicted one-step to three-step ahead traffic flow prediction result is shown in Table-5.1.

TABLE-5.1, Traffic flow prediction on CTW road section

Step	Cell	Trained Instance	Traffic Speed			Traffic Flow		
			MRE	MAE	RMSE	MRE	MAE	RMSE
1	26+26	900	7.84	0.34	0.43	12.18	17	24
2	25+25	900	7.78	0.35	0.46	12.14	17	24
3	24+24	900	7.97	0.35	0.41	14.09	19	27

According to Table-5.1, the model's mean relative error range is 7.78 %~ 7.97% for traffic speed and 12.14% ~ 14.09% for traffic flow prediction through 1-step to 3-steps ahead. The mean relative error is small and stable in traffic speed and comparatively 4% -6% higher than the predicted traffic flow. This is because, while mapping predicted traffic speed into predicted traffic flow using time-dependent correlation equations, the traffic flow deferred with the same traffic speed and traffic flow varying over time. Sometimes observed the weak correlations several times. Subsequently, the model-1 traffic flow prediction graph is shown in Fig-5.1 to understand the flow prediction accuracy over time. The graph indicates a maximum three-step or fifteen-minute ahead traffic flow prediction for five minutes interval of evening data. This road section handles more traffic during evening hours due to the number of vehicles are returned at this time. Sometimes, model-1 underfitting and overfitting due to time-dependent correlation equations while predicting traffic flow. Also, the unwanted traffic flow causes model traffic flow prediction accuracy to slightly degrade in some steps. Moreover, this road is interrupted by a train signal. The model prediction accuracy is comparatively insignificant in both speed and flows prediction and the maximum traffic flow prediction accuracy is achieved a maximum of 88% approximate at the evening hour data.

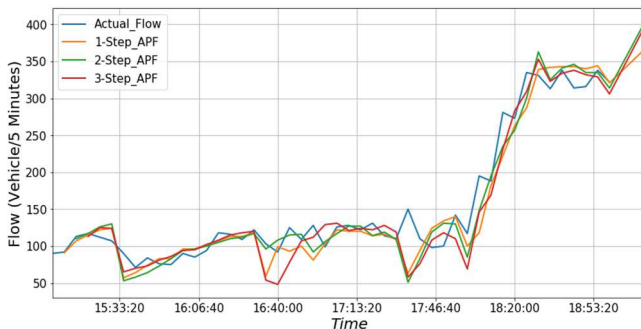


Fig-5.1, CTW traffic flow prediction

Consequently, model-2 is simulated for similar multi-step ahead traffic flow prediction on the CTN road section and its consider a more demanding or busy road due to the highest numbers of traffic flow existing each day. In the morning traffic flow is higher than the evening traffic. Moreover, the sum of the two roads section traffic flow is considered for analysis. This (CTN) road section's one to three steps ahead traffic speed and traffic flow prediction investigation result is shown in Table-5.2. According to Table-5.2, the model predicted mean relative error (MRE) range 8.99%~9.34% during predicted traffic speed and 8.25% ~ 8.82% for traffic flow prediction.

TABLE-5.2, Traffic flow prediction on CTN road section

Step	Train	Cell	Time Interval	Traffic Speed			Traffic Flow		
				MRE	MAE	RMSE	MRE	MAE	RMSE
1	1801	60+60	5	9.32	0.65	0.82	8.82	54	81
2	1801	56+56	5	9.34	0.64	0.82	8.76	54	81
3	1801	53+53	5	8.99	0.60	0.77	8.25	52	78

Model traffic flow prediction error differs approximately 1% from predicted traffic speed. The model traffic speed prediction and traffic flow prediction error is stable in each step and also more closely predicted in each step. The model flow prediction accuracy is higher compared to speed prediction because this road section is uninterrupted and has stable flow all times from morning to evening. Moreover, maximum time-dependent correlation equations are strong relation. The maximum fifteen-minute ahead traffic flow prediction accuracy is 91% approximate which is high for urban road networks. Model three-step ahead prediction performance is higher from morning to evening hours shown in Fig-5.2. Three steps or fifteen-minute ahead prediction status graph indicates that every step's prediction is much closed to the next step prediction at all times. At the morning hour, the model over predicted the traffic flow due to the higher number of traffic and frequently changed the pattern whereas all other times in a day, flow prediction is stable even three steps ahead of prediction.

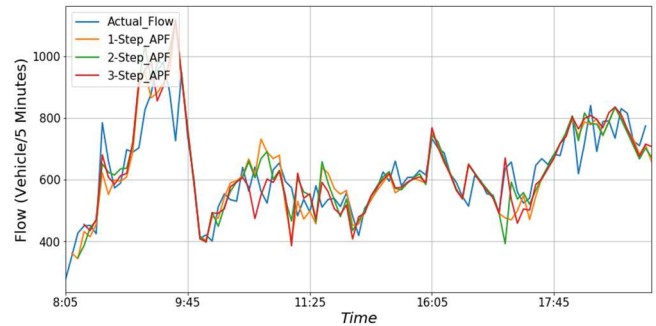


Fig-5.2, CTN traffic flow prediction

The final traffic flow prediction experiment is performed by the developed model-3 on the army stadium road section (STN) for multi-step traffic flow prediction. The STN road section average traffic speed range is 25.41 miles/hr. ~ 32.33mile/hr. in a day and it's considered for free-flow or freeway traffic in the urban road network in this study. This research investigated the proposed model-3 prediction performance on the freeway or free-flow traffic (high speed) and it's experimental multi-step ahead prediction result is shown in Table-5.3. The developed model multi-step prediction mean relative error range 2.76%~ 2.94% for

average traffic speed and 8.82%~9.02% for traffic flow. The developed model average prediction accuracy is higher and stable than other's road sections for multi-step ahead traffic speed prediction.

TABLE-5.3, Traffic flow prediction on STN road section

Step	Train	Cell	Time Interval	Traffic Speed			Traffic Flow		
				MRE	MAE	RMSE	MRE	MAE	RMSE
1	1801	82+82	5	2.87	0.83	1.03	8.82	27	38
2	1801	75+75	5	2.76	0.79	1.00	9.02	27	38
3	1801	69+69	5	2.94	0.83	1.05	8.9	27	36

Model-3 traffic flow prediction accuracy is also 91% approximate which is high comparatively for three-step ahead traffic flow prediction. Moreover, model mean absolute error and root mean square is stable and small. Model prediction error defers 5% from speed to traffic flow mapping. So that, we can say that the model overall prediction performance is higher and stable while investigating freeway traffic flow. The model five-minute interval traffic flow of multi-step ahead prediction graph is shown in Fig-5.3. The model is three-step ahead of traffic flow prediction and very close to each other in each step.

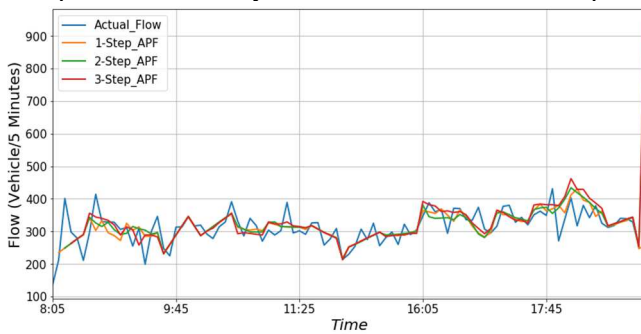


Fig-5.3, STN traffic flow prediction

VI. CONCLUSION

This study investigated three categories of traffic flow prediction are performed by separating three types of road sections like low demand, high demand and free-flow traffic. Developed models traffic flow prediction error range is 8.25%~14.09% for multi-step ahead prediction using five-minute time interval data. The little bit demand road (CTW) prediction error range is 12%~14%, error range is 8%~9% at high demand (CTN) and error range is 8%~9% at freeway traffic (STN) respectively. The CTW road section prediction accuracy is smaller compared to others due to uncertain traffic flow created by motor vehicles and cars whereas the other two road sections traffic flow is stable all day. Also, time-dependent correlation causes slightly increasing error while mapped predicted speed to traffic flow. Moreover, these three model's prediction accuracy will be improved by employing 24 hours more dataset and more field level flow data. Models prediction accuracy is high and stable while investigating identical and freeway flow traffic. The proposed model prediction accuracy is high even in rush hour at five minutes intervals of sequential data. The proposed stacked LSTM deep

network model is suitable for processing sequential data or time-series data with high accuracy.

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